

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA1304

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 10%

QUESTIONS: 10
Data Interpretation

Total Marks:100

Code of Conduct:

*I **Shaik Mohammed Waseem Rayan (192373004)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Shaik Mohammed Waseem Rayan

SIMATS ENGINEERING ASSESSMENT TOOLS

Given Data (Grammar)

Consider the CFG:

$S \rightarrow aSb \mid ab$

Q1. Language Identification

Based on the given grammar, which of the following strings belong to the language? A) ab

B) aabb

C) aaabbb

D) abb

ANSWER:

The grammar generates strings with equal numbers of as and bs, with all as before the bs:

- $S \rightarrow ab \rightarrow ab$
- $S \rightarrow aSb \rightarrow a(ab)b \rightarrow aabb$
- $S \rightarrow aSb \rightarrow aaSbb \rightarrow aa(ab)bb \rightarrow aaabbb$

Therefore, the correct answers are:

- ☒ A) ab
- ☒ B) aabb
- ☒ C) aaabbb

Q2. Pattern Recognition

From the grammar, identify the correct pattern:

A) $a^n b^n$

B) $(ab)^n$

C) $a^n b^m$

D) $(a+b)^*$

Answer: A) $a^n b^n$

Explanation:

The grammar $S \rightarrow aSb \mid ab$ generates strings with **equal numbers of as followed by equal numbers of bs**.

Examples: ab, aabb, aaabbb

Therefore, the pattern is $a^n b^n$.

Q3. Derivation Analysis

Which of the following is a valid derivation for aabb?

A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

B) $S \rightarrow ab \rightarrow aabb$

C) $S \rightarrow aSb \rightarrow ab$

D) $S \rightarrow aaSbb \rightarrow aabb$

Answer: A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

Short Explanation:

Starting from S:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow aa(ab)bb \rightarrow aabb$

Therefore, **A is the valid derivation.**

Q4. Missing Step Identification

Fill in the missing step:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

A) ab

B) S

C) ϵ

D) aSb

Answer: A) ab

Explanation:

Using $S \rightarrow ab$:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow aa(ab)bb \rightarrow aabb$

So the missing step is **ab**.

Q5. Parse Tree Interpretation

How many levels are there in the parse tree for string aaabbb?

A) 2

B) 3

C) 4

D) 5

Answer: D) 5

Explanation:

For aaabbb:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow aaSbbb \rightarrow aaabbb$

The parse tree has **5 levels**.

SIMATS ENGINEERING ASSESSMENT TOOLS

Consider:

$S \rightarrow SS \mid a$

Q6. Ambiguity Check

This grammar is:

A) Unambiguous

B) Ambiguous

C) Regular

D) Deterministic

Answer: B) Ambiguous

Explanation:

The grammar $S \rightarrow SS \mid a$ is **ambiguous** because some strings can have **more than one parse tree**.

Q7. String Generation

Which string can be generated?

- A) a
- B) aa
- C) aaa
- D) All of the above

Answer: D) All of the above

Explanation:

Using $S \rightarrow SS \mid a$, we can generate:

- $a \rightarrow$ directly from $S \rightarrow a$
- $aa \rightarrow S \rightarrow SS \rightarrow aa$
- $aaa \rightarrow S \rightarrow SS \rightarrow aSS \rightarrow aaa$

So, **all three strings can be generated**.

Q8. Interpretation of Ambiguity

Ambiguity means:

- A) One string $\rightarrow$ one parse tree
- B) One string $\rightarrow$ multiple parse trees
- C) No derivation
- D) Infinite symbols

Answer: B) One string $\rightarrow$ multiple parse trees

Explanation:

A grammar is **ambiguous** when a single string can have **more than one valid parse tree** or derivation.

Final Answer: B) One string $\rightarrow$ multiple parse trees.

Given Data (Grammar 1)

$S \rightarrow 0S1 \mid 01$

Q9. Derivation Interpretation

Which of the following derivations correctly generates the string 0011? A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$

- B) $S \rightarrow 01 \rightarrow 0011$
- C) $S \rightarrow 0S1 \rightarrow 01$
- D) $S \rightarrow 00S11 \rightarrow 0011$

Answer: A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$

Explanation:

Using $S \rightarrow 01$ in the final step:

$S \rightarrow 0S1 \rightarrow 0(01)1 \rightarrow 0011$

Final Answer: A.

Given Data (Grammar 2)

$$S \rightarrow aS \mid Sb \mid ab$$
Q10. Structural Analysis

Which of the following strings can be generated by the grammar? A) ab

B) aab

C) abb

D) All of the above

Answer: D) All of the above

Explanation:

The grammar $S \rightarrow aS \mid Sb \mid ab$ generates strings with one or more as followed by one or more bs.

- ab ☒
- aab ☒ $\rightarrow S \rightarrow aS \rightarrow a(ab)$
- abb ☒ $\rightarrow S \rightarrow Sb \rightarrow (ab)b$

Final Answer: D) All of the above.

SIMATS ENGINEERING ASSESSMENT TOOLS

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application

Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand